



Esterification of Sago Dregs Bio-Oil Using Zeolite Modified MgO for Biofuel Applications

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Abstract: Bio-oil from lignocellulosic biomass pyrolysis cannot be applied as biofuel because it is generally corrosive due to its high organic acid content. The organic acid content of heavy fractions can be reduced by fractionation and by esterification with alcohol to form ester compounds. This study aims to produce high-quality fuel by optimizing the bio-oil: methanol ratio using a magnesium metal oxide modified H-zeolite catalyst (HAAZ/MgO) to convert sago dregs bio-oil. This study was carried out in several stages: pyrocatalytic sago dregs were heated to 350–500 °C, then the bio-oil was filtered and fractionally distilled at 91–110 °C. HAAZ/MgO catalyst was successfully synthesized according to Fourier transform infrared spectroscopy (FTIR) characterization, showing absorption at 3300–3700 cm⁻¹, the emergence of hydroxyl group (-OH) stretching vibrations originating from silanol groups (Si-OH) and Brønsted acid sites (Si-OH-Al), 1641–1649 cm⁻¹ as H-O-H bending vibrations, 1053–1223 cm⁻¹ asymmetric stretching vibrations of Si-O-Si and Si-O-Al bonds, and 1350–1450 cm⁻¹ indicating the presence of MgO-zeolite. The X-ray diffraction (XRD) spectrum of HAAZ/MgO shows diffraction peaks at $2\theta = 20.86^\circ$, 25.67° , 26.65° , and 27.74° . The presence of MgO does not damage the HAAZ structure and is evenly dispersed. The fractionated distillate was esterified by reflux at 65 °C. at the ratio of bio-oil distillate to methanol (1:6, 1:8, and 1:10) using HAAZ/MgO catalyst. The esterified biofuel showed the best yield at a 1:10 ratio, with $72.22 \pm 1.11\%$ (v/v). The esterification process demonstrated the HAAZ/MgO catalyst's good performance, yielding dimethyl and methyl esters. In addition, the physicochemical properties of bio-oil, including pH, viscosity, and API gravity, increased significantly after esterification, while water content, density, specific gravity, and viscosity decreased. Meanwhile, the higher heating value (HHV) of the esterified biofuel increased from 43.55 to 45.15 MJ/kg. Improvements in these parameters indicate that the esterification process plays an important role in enhancing biofuel quality, making it a feasible and efficient renewable energy source.

Keywords: Bio-oil; Sago dregs; Zeolite; MgO; Pyrocatalytic; Fractionation; Esterification; Biofuel

1 Introduction

The limited reserves of fossil fuels encourage the development of alternative renewable energy as a substitute or complement to conventional fuels [1]. Currently, biomass-based biofuels from agricultural waste are the focus of research because they are carbon-neutral and abundantly available. In this study, sago dregs, which are abundant in Southeast Sulawesi, are used as a raw material for biofuel production due to their high lignocellulose content, namely 38% cellulose, 23% lignin, and 27% hemicellulose [2]. The high lignocellulose content of sago dregs has great potential to be developed as a renewable energy source through thermochemical conversion processes such as pyrolysis to produce bio-oil.

Bio-oil from lignocellulosic biomass pyrolysis contains various oxygenate compounds, such as carboxylic acids, aldehydes, ketones, phenols, furfurals, and other aromatic compounds. The high oxygen, water, and viscosity

content, and the presence of acidic compounds with low pH and low higher heating value (HHV), make bio-oil corrosive [3–5]. This imposes limitations on the storage, handling, and direct use of bio-oil as a fuel. In addition, the presence of reactive compounds such as aldehydes and phenols can trigger further reactions during storage, leading to increased viscosity, phase separation, and degradation of bio-oil quality [6]. The presence of solids and alkali metals can also cause problems with combustion, clogging, and deposition in energy equipment [7]. Therefore, a quality improvement process is needed to improve its compatibility during storage and enable further processing into a more stable, usable liquid fuel. One method to improve the quality and stability of bio-oil is esterification.

Esterification has been widely reported as a method to improve bio-oil quality, especially by reducing its acid content by converting carboxylic acids or free fatty acids (FFA) into esters with alcohols such as methanol. This process not only reduces acidity but also increases the HHV, with reaction performance strongly influenced by the catalyst type [6, 8]. Conventionally, homogeneous base catalysts (KOH and NaOH) [9, 10] and acids (H_2SO_4 and HCl) [11] have been widely applied due to their high activity. However, their use is limited due to sensitivity to water and FFA, as well as the difficulty of separation in the final product. Therefore, researchers' attention has begun to shift toward developing heterogeneous catalysts that are more stable, recyclable, and environmentally friendly. Among them, single-metal oxides such as CaO, SrO, and magnesium oxide (MgO) [8, 12, 13] have been studied as catalysts in the transesterification of vegetable oil with methanol and have the potential to be effective heterogeneous catalysts.

The development of heterogeneous catalysts based on natural zeolites has become a focus of recent research. In industrial applications, efficiency and productivity are key, so the development of heterogeneous catalysts based on natural materials with high catalytic activity offers significant advantages for biodiesel production via esterification reactions. Zeolite is a crystalline material in the form of hydrated aluminosilicates composed of tetrahedral units (SiO_4^{4-}) and alumina (AlO_4^{5-}), which has been widely studied as a catalyst, especially in the process of improving the quality of bio-oil [14]. This material has a good porous structure and has the potential as a catalyst for bio-oil esterification, as indicated by the presence of Brønsted acid sites on the zeolite surface that can be modified to accommodate the complex composition and characteristics of bio-oil during the esterification process [15, 16].

Surface modification of zeolite with basic metal oxides, such as MgO, can improve the performance of zeolite catalysts in bio-oil esterification reactions by forming bifunctional acid-base catalyst systems. Zeolite provides Brønsted acid sites that play a role in the activation of carboxylic groups, while MgO contributes as basic sites that can facilitate alcohol interactions during the esterification process, so that the combination of acid sites on zeolite and basic sites of MgO can effectively enhance the conversion of organic acid compounds into esters. However, the use of natural zeolite modified with MgO as a heterogeneous catalyst for the esterification of sago dregs bio-oil is still relatively limited in the literature. Therefore, further studies are needed to evaluate the activity and potential of MgO-modified zeolite catalysts in bio-oil esterification reactions. The focus of this research is on the esterification of sago dregs bio-oil with varying mass ratios of bio-oil to methanol, by studying the esterification behavior of organic acids in bio-oil, especially acetic acid, using methanol and MgO-modified zeolite as heterogeneous catalysts.

2 Experimental Section

2.1 Materials

Sago dregs waste obtained from the Andora sago processing factory in Konawe Regency, Southeast Sulawesi, Indonesia, was used as a biomass raw material. The sago dregs were cleaned of impurities, then naturally dried at 29 ± 1 °C for 3–4 days. The chemicals used included zeolite, MgO, methanol (CH_3OH), sulfuric acid (H_2SO_4), hydrochloric acid (HCl), hydrofluoric acid (HF), ammonium chloride (NH_4Cl), ammonia (NH_3), distilled water (H_2O), phenolphthalein indicator (PP), sodium hydroxide (NaOH) 0.1 N, and magnesium sulfate (MgSO_4). All chemicals used were reagents of analytical-grade purity.

2.2 Synthesis of Magnesium Metal Oxide Modified H-Zeolite Catalyst Catalyst

Zeolite with a particle size of 63 μm was sequentially activated using a combination of chemical and physical methods. Chemical activation was carried out by soaking the sample in 1 M HCl for 24 hours, then filtering and washing with distilled water until the pH was neutral. Physical activation was performed by heating in an oven at 150 °C for 3 hours, followed by calcination at 300 °C for 3 hours to produce the H-zeolite catalyst (HAAZ) [3]. Activated H-zeolite (e.g., HAAZ) was modified by impregnation with MgO. The mixture was stirred continuously at ± 70 °C for 1 hour. The solid obtained by heating HAAZ/MgO in an oven at 105 °C for 2 hours was used to prepare the HAAZ/MgO catalyst. Fourier confirmed the formation and functional groups of the catalyst using transform infrared spectroscopy (FTIR), using a Shimadzu IR Prestige-21 instrument over the wavenumber range of 500–4000 cm^{-1} . Meanwhile, crystallinity was characterized by X-ray diffraction (XRD) using an Aeris diffractometer with Cu $\text{K}\alpha$ radiation (40 kV, 15 mA) over a 2θ range of 5°–90°.

2.3 Pyrocatalytic for Bio-Oil Production

The production of sago dregs bio-oil was carried out through fast pyrolysis, with the addition of a zeolite catalyst of 5% (w/w) to the mass of sago dregs. The pyrolysis process was carried out in a cylindrical semi-batch reactor with a 6 cm diameter and a 25 cm height. Thermal decomposition occurred over a temperature range of 350–500 °C, with an initial system temperature of 30 ± 2 °C and a heating rate of 5 °C/s in the absence of oxygen. Pyrolysis experiments at 500 °C were performed five times to ensure reproducibility. The pyrocatalytic gas was condensed to obtain bio-oil, which was then filtered through Whatman paper to separate the tar. The resulting bio-oil was stored in a dark glass bottle at ± 4 °C to prevent re-polymerization. The pyrocatalytic process produced 3 main products: bio-oil, biochar, and tar. The yields of these products were calculated using the gravimetric method, as defined in Eqs. (1) and (2). Meanwhile, the weight of the gas product is calculated based on the difference between the total weight of the initial raw material (W_0) and the weight of the identified solid product W_x .

$$\text{Yield}_{(\text{Bio-oil})}(\%) = \frac{\text{Weight}_{(\text{Bio-oil})}}{w_0} \times 100\% \quad (1)$$

$$\text{Yield}_{(\text{Biochar})}(\%) = \frac{\text{Weight}_{(\text{Biochar})}}{w_0} \times 100\% \quad (2)$$

2.4 Purification of Bio-Oil

Bio-oil was purified by fractional distillation. The distillation process was carried out at 91–110 °C for ± 3 hours on 100 mL of bio-oil. The collected distillate consisted of a lighter bio-oil fraction, separated from tar and other heavy elements, with physical characteristics ranging from clear to yellowish. The distillate bio-oil was then stored in an Erlenmeyer flask to prevent contamination and sample stability. This method is a modification of previous research by Mashuni et al. [3] In this study, the distillate bio-oil was not only analyzed for quality and chemical composition but also subjected to further upgrading through esterification to improve its physicochemical properties.

2.5 Catalytic Esterification of Bio-Oil

The bio-oil used in the esterification process is a purified bio-oil obtained by distillation and contains carboxylic acids, as confirmed by GC-MS analysis of the distillate (Table 1). The esterification process aims to convert the alcohol and carboxylic acid present in bio-oil into esters using acid or base catalysts. In this study, esterification was carried out using reflux at a temperature of 65 °C for 2 hours with variations in the ratio of bio-oil: methanol, namely 1:6; 1:8 and 1:10, the catalyst used was a magnesium metal oxide modified H-zeolite catalyst (HAAZ/MgO) at a ratio of 4:1 with a concentration of 10% (w/v) in the volume of bio-oil: methanol. The water content of the bio-oil produced by the esterification process was removed by adding MgSO_4 , and the water content was determined gravimetrically. This method is a modification of the research by Jahiding et al. [17] and Kadarwati et al. [18]. All stages in this research are shown in Figure 1.

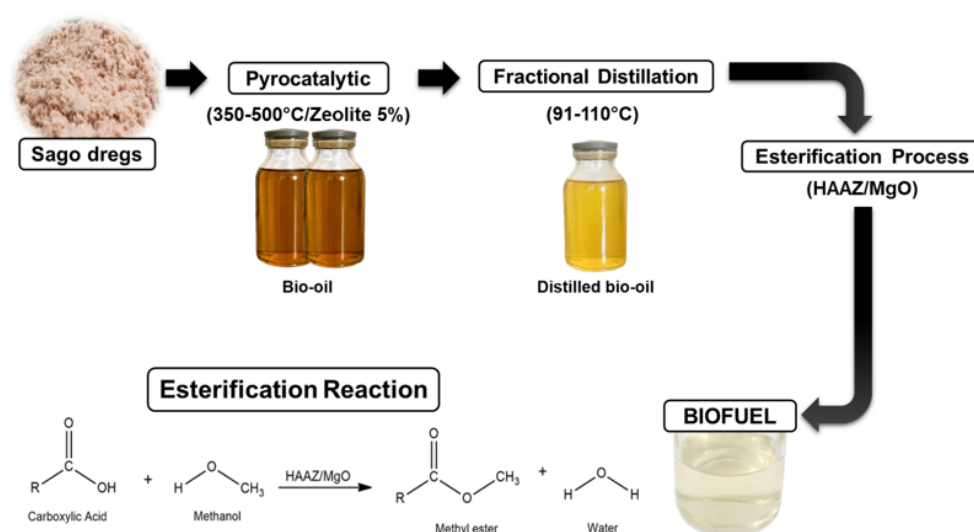


Figure 1. Schematic flow diagram of the biofuel production process from esterified sago dregs

Note: HAAZ = activated natural zeolite.

Table 1. Chemical components of bio-oil, distillate, esterified biofuel

Chemical Compounds	Molecular Formula	SI	Area (%)		
			Bio-Oil	Distillate	Esterification 1:10
Carboxylic acid					
Acetic acid	C ₂ H ₄ O ₂	93	28.23	58.26	-
Propanoic acid	C ₃ H ₆ O ₂	84	8.30	-	-
Octadecadienoic acid	C ₁₈ H ₃₂ O ₂	73	-	39.55	-
11-octadecenoic acid	C ₁₈ H ₃₄ O ₂	70	-	-	18.50
13-docosenoic acid	C ₂₂ H ₄₂ O ₂	69	-	-	20.30
Ester					
Dimethyl ester	C ₁₇ H ₁₉ NO ₄	79	-	-	27.73
Acid methyl ester	C ₂₅ H ₂₅ N ₃ O ₇	70	-	-	23.30
Aldehyde					
Acetaldehyde	C ₂ H ₄ O	95	1.69	-	-
Propanal	C ₃ H ₆ O	93	21.30	-	-
Ketones					
2-propanone	C ₃ H ₆ O	98	14.33	28.91	-
Cyclopentanone	C ₅ H ₈ O	89	2.23	-	-
1-hydroxy-2-butanone	C ₄ H ₈ O ₂	83	0.45	-	-
5-hydroxyheptan-2-one	C ₇ H ₁₄ O ₂	94	0.94	-	-
2-cyclopenten-1-one	C ₅ H ₆ O	98	2.34	1.66	-
Alcohol					
1-octanol	C ₈ H ₁₈ O	78	-	-	12.15
Phenolic					
Phenol	C ₆ H ₆ O	97	7.38	2.87	-
Furan					
5-methylfurfural	C ₆ H ₆ O ₂	93	3.20	-	-
Heterocyclic					
Pyrazole	C ₃ H ₄ N ₂	88	9.24	-	-

2.6 Analysis of GC-MS

The chemical composition of bio-oil, bio-oil distillate, and biofuel was determined using a GC-MS instrument (Shimadzu QP2010SE). The injector temperature was set to 280 °C, with a split ratio of 1:10, using pure helium as the carrier gas. The oven program started with an initial temperature of 50 °C for 5 minutes, then heated to 280 °C at 5 °C/minute, and maintained the final temperature for 5 minutes. The mass spectrometer used electron ionization (EI) with an ionization energy of 70 eV, a scanning speed of 1 scan/second, a mass range of m/z 30–500 amu, and an ion source temperature of 230 °C. Identification of chemical compounds involved comparing the retention times and fragmentation patterns of detected peaks with standards from the National Institute of Standards and Technology (NIST) mass spectral library.

2.7 Analysis of Physicochemical

Physicochemical analyses of bio-oil, bio-oil distillate, and biofuel were performed for water content, pH, viscosity, density, specific gravity (SG), API gravity, and HHV. Water content was determined by the gravimetric method, and pH was measured with a pH meter (WT61) calibrated with buffer solutions 4, 7, and 10. Viscosity was measured using an Ostwald viscometer by determining the fluid flow rate through the capillary. Density was determined using a pycnometer by comparing the weight of distilled water and bio-oil at the same temperature. The SG and API gravity were measured according to IS 1448 [P:32]:1992 and ASTM D1298-12B (2017) standards. The SG value was calculated based on the ratio of sample density to water density, while API gravity was calculated using a standard relationship with SG. As given in Eqs. (3) and (4), respectively, where SG is the specific gravity of the sample relative to water at the standard reference temperature of 15.6 °C.

$$\text{Specific gravity(SG)} = \frac{\rho_{\text{sample}}}{\rho_{\text{water}}} \quad (3)$$

$$\text{APIgravity} = \frac{141.5}{\text{SG}} - 131.5 \quad (4)$$

The HHV was calculated using an empirical correlation based on API Gravity, which is widely used for petroleum-derived fuels and similar hydrocarbons. As reported in McGraw-Hill's Engineering Companion [19], and expressed in Eq. (5), where HHV is expressed in Btu/lb and API gravity. The calculated HHV was subsequently converted into SI units (MJ/kg) using a standard conversion factor.

$$\text{HHV} = 18,650 + 40(\text{API Gravity}-10) \quad (5)$$

The determination of acid number according to SNI 04-7182-2015 uses the alkalimetric titration method. Five grams biofuel sample is dissolved in ethanol, and phenolphthalein indicator is added. The solution is then titrated with 0.1 N NaOH until a stable color change occurs at the titration end point. The volume of NaOH used is recorded and used to calculate the acid number value, using Eq. (6):

$$\text{Acid number} = \frac{V_{\text{NaOH}} \times N_{\text{NaOH}} \times \text{BMsample bio-oil}}{\text{Weight sampel}} \times 100\% \quad (6)$$

3 Result and Discussion

3.1 Fourier Transform Infrared Spectroscopy Analysis of Magnesium Metal Oxide Modified H-Zeolite Catalyst Catalyst

FTIR analysis of HAAZ and HAAZ/MgO catalysts (Figure 2) shows several typical peaks of aluminosilicate materials that identify the characteristics of the catalyst functional groups. In the range of 3300–3700 cm^{-1} , the appearance of stretching vibrations of hydroxyl groups -OH originating from silanol groups (Si-OH) and Bronsted acid sites (Si-OH-Al). The HAAZ catalyst shows a broader peak due to the heterogeneous distribution of -OH groups. Conversely, HAAZ/MgO shows a sharper peak, indicating a more homogeneous distribution of -OH groups due to the interaction of cations (Mg^{2+}) and -OH groups on the surface [20]. This MgO group accelerates esterification, especially in the conversion of alcohols into esters with carboxylic acids.

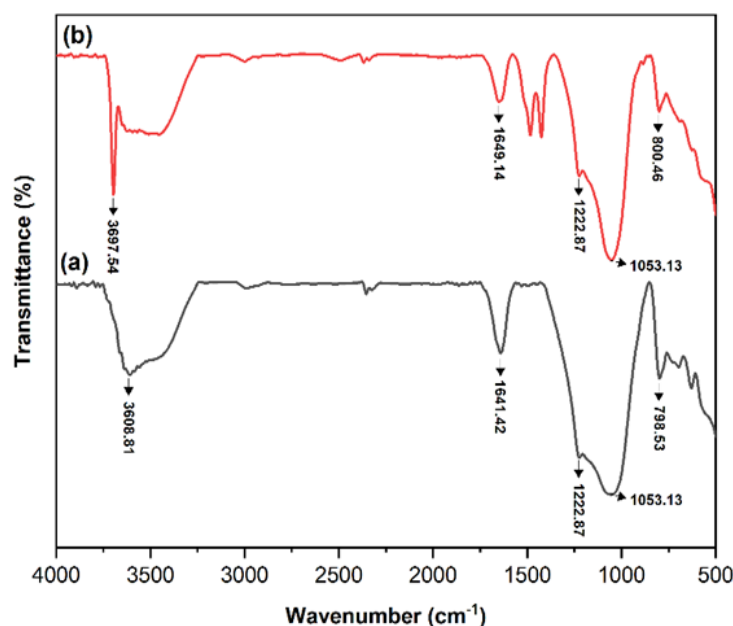


Figure 2. IR spectrum of catalysts (a) HAAZ and (b) HAAZ/MgO

Note: HAAZ = activated natural zeolite.

Another peak at 1641–1649 cm^{-1} is the H-O-H bending vibration of adsorbed water molecules, which is characteristic of zeolite hydrophilicity [20, 21]. There is a prominent peak in the region of 1053–1223 cm^{-1} with the emergence of asymmetric stretching vibrations of Si-O-Si and Si-O-Al bonds, and 1350–1450 cm^{-1} indicates the presence of MgO-zeolite. However, in the range of 700–500 cm^{-1} , there is a slight shift of the peaks associated with the symmetric stretching vibrations of the zeolite structure. This indicates that MgO impregnation does not significantly alter the basic structural framework of the aluminosilicate zeolite.

3.2 X-ray Diffraction Analysis of Katalisator Magnesium Metal Oxide Modified H-zeolite Catalyst Catalyst

Catalyst performance is crucial for determining the efficiency and yield of the esterification reaction [22, 23]. The XRD diffraction pattern of activated zeolite (HAAZ) and MgO-modified HAAZ (HAAZ/MgO) shows peaks

identified at 2θ values (Figure 3). The XRD diffraction of HAAZ (Figure 3a) shows peaks at $2\theta = 20.82^\circ$, 25.69° , 26.61° , and 27.64° , which are consistent with the Joint Committee on Powder Diffraction Standards (JCPDS) file 700232 [24]. The diffraction spectrum of HAAZ/MgO in this study shows peaks at $2\theta = 20.86^\circ$, 25.67° , 26.65° , and 27.74° (Figure 3b).

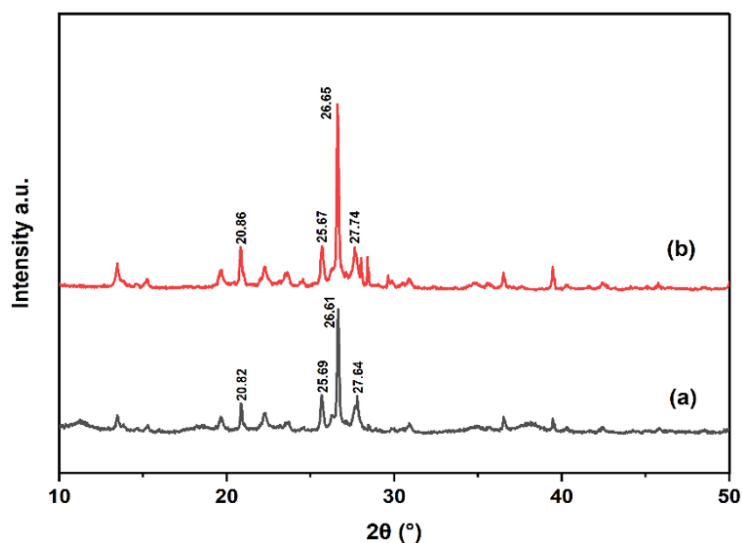


Figure 3. Diffraction patterns of (a) HAAZ and (b) HAAZ/MgO

Note: HAAZ = activated natural zeolite.

XRD analysis shows that the surface structure of HAAZ remains stable and undergoes no significant changes during MgO impregnation. This indicates that MgO modification does not cause damage to the HAAZ crystal lattice. These peaks indicate that MgO is well dispersed in the HAAZ matrix, thus providing the expected catalytic properties. The absence of significant new peaks in the diffraction pattern may be due to the added MgO content being below the measurement limit, or to MgO being evenly dispersed in HAAZ. In addition, the peaks at $2\theta = 36.91^\circ$, 42.12° , and 42.91° are the main diffraction peaks of MgO, which has a polymorphic structure according to JCPDS file 45-0946 [25]. The interaction between HAAZ and MgO indicates that HAAZ is a stable support for MgO, thereby maintaining its structural stability and catalytic activity.

3.3 Yield Product of Bio-Oil

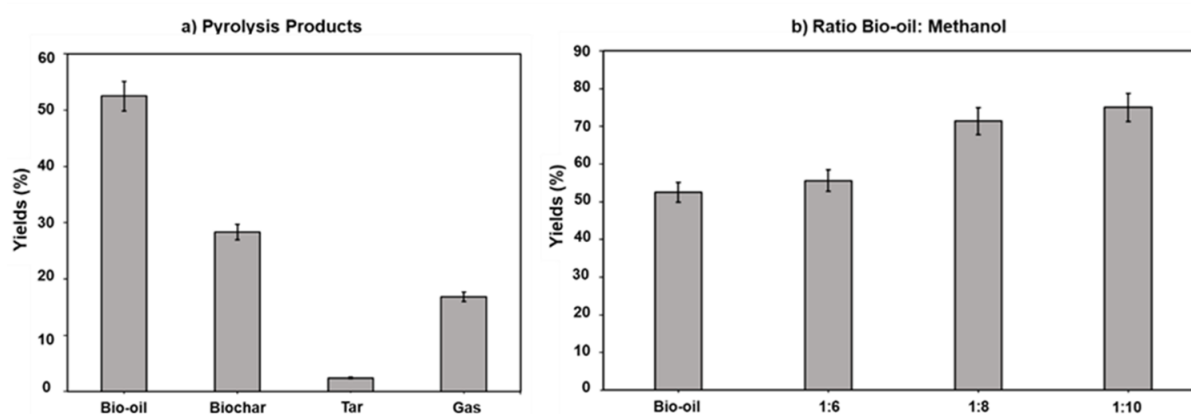


Figure 4. (a) Pyrolysis product yields and (b) esterification of bio-oil: methanol at different ratios

Bio-oil production in this study was carried out through pyrolysis with the help of a zeolite catalyst (pyrocatalytic). The pyrolysis process was carried out continuously, starting at 350°C and reaching a maximum of 500°C . The reaction produced several products, including bio-oil, tar, biochar, and gas (Figure 4). Bio-oil became the dominant fraction with an average yield of 52.48 ± 0.01 wt%. The results of this study are close to those reported by Liu et al. [17], who obtained 52.87% at 500°C with zeolite addition. This indicates that the use of a temperature of 500°C and the addition of zeolite effectively encourages the decomposition of lignocellulosic components, so that most of the

sago dregs mass is successfully converted into bio-oil, a liquid rich in organic compounds, and has the potential to be used as an alternative fuel [26].

The bio-oil purification process yielded a pure bio-oil fraction with a $62.3 \pm 0.3\%$ yield. This value indicates that more than half of the bio-oil volume was successfully separated into a higher-purity distillate. This percentage indicates the effectiveness of the distillate in separating light fractions that have higher commercial value from bio-oil. This purification success plays an important role in improving the quality of bio-oil, making it more suitable for use in further applications. The light fraction obtained from distillation also has a HHV and better stability [26, 27]. However, the bio-oil from distillation still does not meet fuel criteria, as this process does not significantly reduce the organic acid content. This condition is confirmed by the increase in acetic acid concentration, as shown in Table 1. Given these limitations, an upgrading stage is needed to reduce the organic acid content and improve the physicochemical properties of the bio-oil through esterification.

The esterification process converts carboxylic acids into more stable ester compounds. This study showed that bio-oil yields varied greatly with the bio-oil-to-methanol ratio, with esterification product yields of $66.92 \pm 0.7\%$, $70 \pm 0.91\%$, and $72.22 \pm 1.11\%$ at ratios of 1:6, 1:8, and 1:10 [28, 29]. The esterification reaction used in bio-oil purification is the Fischer reaction, which involves the reaction of carboxylic acids with alcohols to form esters. This reaction is important because esters exhibit better physical and chemical properties than the initial carboxylic acids, thereby improving the quality of the resulting bio-oil. The decrease in yield at higher methanol ratios in this study can be explained by the references [30], which states that the use of excess alcohol can reduce the efficiency of the esterification reaction, shift the reaction equilibrium, or make separation of the ester products more difficult. Therefore, our results emphasize the importance of selecting the right ratio to maximize esterification yield and improve bio-oil quality.

3.4 Chemical Composition of Bio-Oil

Pyrocatalytic sago dregs at a temperature of 350–500 °C produce bio-oil rich in various chemical compounds, as shown in Table 1. Based on the analysis results, several chemical compounds have been detected in bio-oil, including groups of carboxylic acids, ketones, aldehydes, pyrazoles, phenolics, and furfural compounds. Research by Bridgwater [31] on the decomposition of sago dregs at 400–600 °C shows that pyrolysis temperature strongly influences the types and distributions of chemical compounds, with the dominant compounds being carboxylic acids, ketones, phenolics, and furfural. Meanwhile, Yanti et al. [32] at a pyrolysis temperature of 250–500 °C produced 57.55% of the dominant phenolic compounds, with other compounds, such as aromatics, hydrocarbons, alcohols, and furfural, in lower amounts. The pyrolysis temperature strongly influences the chemical composition, thereby determining the compounds formed during biomass decomposition. Carboxylic acid compounds from bio-oil are formed in significant quantities due to the breakdown of cellulose and hemicellulose at low to moderate pyrocatalytic temperatures [33]. Ketone compounds in bio-oil are formed due to the breakdown of acetyl groups in lignocellulose components and the decomposition of simple sugars in biomass. Meanwhile, phenolic compounds are produced from the breakdown of lignin in biomass.

The boiling point of the purification process refers to the vapor pressures of the heavy and light fractions that affect distillation. Compounds with lower boiling points and higher volatility usually increase in the distillation fraction. Conversely, compounds with higher boiling points or that are chemically unstable may be less detectable or exhibit a decrease in concentration [33, 34]. For example, in this study, volatile compounds with low boiling points, including 2-propanones and acetic acid, are distilled in the 91–100 °C range [35]. This shows an increase in the concentration of these two compounds in the distilled bio-oil, with acetic acid identified as the dominant component at 58.26%, indicating a high proportion of organic acids in the fraction. The effectiveness of the distillate in separating bio-oil components by boiling point affects the quality and applications of bio-oil. The high acetic acid content also enhances esterification efficiency with methanol, producing esters and water as byproducts.

Esterification is a chemical reaction between alcohol and carboxylic acid that produces ester compounds. The esterification process aims to reduce the amount of free organic acids and increase the HHV of bio-oil. Bio-oil containing strong organic acids (carboxylic acids) can be esterified by the addition of alcohol (methanol). The active sites involved in the esterification of organic acids are usually identified as Brønsted acid sites, and the reaction proceeds through a mechanism similar to that of homogeneous systems [6, 35]. The HAAZ/MgO catalyst facilitated the esterification of carboxylic acids with methanol through a bifunctional acid-base catalytic mechanism, as illustrated in Figures 5 and 6.

The activity of the carboxylic acid carbonyl group occurs when protons (H^+) and Brønsted sites HAAZ protonate carbonyl oxygen ($C=O$), which increases the positive charge on the carbon atom so that carbon becomes more electrophilic and easily attacked by methanol, while reducing the activation energy of the reaction and accelerating the esterification rate. The microporous structure of zeolite provides a pore selectivity effect by limiting the occurrence of side reactions and increasing the selectivity of the desired ester, because without the presence of zeolite, the esterification reaction takes place very slowly [36–38]. Meanwhile, MgO acts as a base component that activates

methanol through the interaction of the base site with the -OH group of methanol, forming methoxide (CH_3O^-) which is a more reactive nucleophile, while adsorbing water as a reaction byproduct that thermodynamically can shift the equilibrium towards the reactant and deactivate the acid site, so that its presence suppresses the inhibitory effect of water and balances the acidity of the reaction by reducing excess acidity, and prevents side reactions such as dehydration and cracking [39, 40]. The synergistic effect of HAAZ/MgO is the main factor that determines the effectiveness of esterification, so that the interaction of the two increases the reaction rate, conversion, and selectivity of the ester, and allows the reuse of heterogeneous catalysts that are more environmentally friendly than conventional homogeneous catalysts such as H_2SO_4 [41, 42].

Dimethyl ester compounds were found in esterified biofuels at 27.73%, while acid methyl esters were 23.30%. The total ester compound content was 51.03%, indicating that the esterification process was effective and significantly improved the physicochemical quality of biofuels, as shown in Table 1. The presence of methyl esters and acid methyl esters in bio-oil plays an important role in improving fuel quality, as esterification can reduce viscosity, acidity, and water content. Bio-oil becomes more stable [43], therefore, esterification and distillation are effective approaches to producing more environmentally friendly fuels with higher combustion efficiency and lower emissions. Distillation removes volatile compounds, and esterification increases ester levels, improving the stability and characteristics of biofuels.

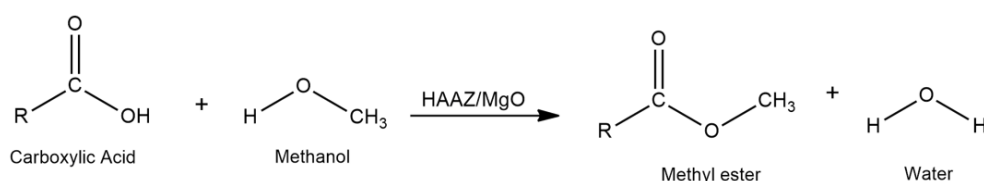


Figure 5. Esterification reaction

Note: HAAZ = activated natural zeolite.

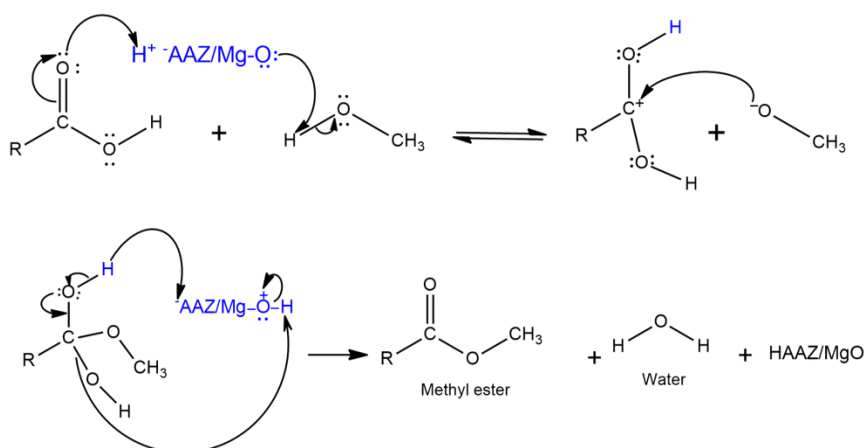


Figure 6. Mechanism of esterification

Note: HAAZ = activated natural zeolite.

3.5 Physicochemical of Bio-Oil

Table 2 shows the physicochemical properties of bio-oil produced from sago waste biomass through pyrocatalytic, distillation, and esterification processes. The results show that the produced bio-oil has characteristics similar to those of conventional fuels, but it still requires further improvements to reach the application stage. Esterification is a chemical process in which acetic acid reacts with methanol to form esters and water, with the help of a HAAZ/MgO catalyst, to accelerate the reaction. The high acetic acid content of the pyrocatalytic bio-oil results in a 28.23% bio-oil content, which increases to 58.26% after distillation. High acetic acid content can accelerate ester formation, reduce acidity, and improve the quality of bio-oil. After esterification, the resulting bio-oil showed improvements in physicochemical properties, including decreases in pH, water content, density, viscosity, and HHV [18, 43]. This process is very important for improving the quality of bio-oil, whose application remains limited due to its corrosivity and high acidity.

Water content plays a role in determining the thermal characteristics of bio-oil. The water content in this study decreased gradually from 0.34% in the bio-oil to 0.33% after purification. Furthermore, in bio-oil that has undergone

esterification with methanol ratios of 1:6, 1:8, and 1:10 (0.23–0.22%). The decrease in water content indicates that the esterification process, aided by the HAAZ/MgO catalyst, not only modifies the chemical composition of the bio-oil but also effectively reduces its water content, thereby improving its quality. High water content is known to reduce the flame temperature, combustion efficiency, and combustion reaction rate. However, in certain amounts, water can increase the pH of bio-oil. Bio-oil shows high acidity (pH 3.71), originating from oxygenate compounds, especially organic acids, resulting from pyrolytic biomass [44, 45]. The distillation process slightly reduces acidity (pH 4.64), suggesting a reduction in volatile acid components. This change is not very significant and serves only as an initial purification stage, but it is not yet effective enough at reducing acidity levels. After esterification, the bio-oil pH increases to 6.24–7.00, indicating the esterification process successfully reduced its acidity. Bio-oil with a high acidity level is corrosive to metal components, which are not ideal for combustion engines [46, 47]. Therefore, increasing the pH during esterification improves the physicochemical properties of bio-oil by reducing its corrosive potential and increasing its suitability as an alternative fuel.

Table 2. Physicochemical properties of bio-oil, distillate, and esterified biofuel

Properties	Bio-Oil	Distillate	Biofuel Esterifikasi			Conventional	
			(1:6)	(1:8)	(1:10)	Gasoline	Diesel
Color	Dark brown	Amber yellow	Light yellow			Light yellow	Light yellow
pH	3.71	4.64	6.24	6.54	7.00	N/A	N/A
Moisture (wt%)	0.34	0.33	0.23	0.23	0.22	0	0
Density (g/cm ³)	0.98	0.97	0.88	0.84	0.84	0.71–0.77	0.82–0.86
Viscosity (cSt)	0.99	0.98	0.93	0.90	0.90	0.3–0.5	2.0–4.0
Specific gravity	0.98	0.97	0.88	0.84	0.84	-	-
API gravity (API)	12.30	13.08	28.32	28.50	28.56	-	-
Higher heating value (MJ/kg)	43.55	43.64	45.07	45.10	45.15	44–47	42–47
Acid number	0.019	0.012	0.01	0.01	0.01	N/A	N/A

The density of bio-oil is generally in the range of 1.1–1.3 g/cm³, which is relatively higher than that of conventional fossil fuels [41, 42]. In this study, the bio-oil density was 0.98 g/cm³ and decreased by only 0.01 g/cm³ after distillation. This indicates that most of the heavy compounds have evaporated during the distillation process [31]. After the esterification process, the density value decreased significantly to 0.88 g/cm³ at a ratio of 1:6 and continued to decrease as the methanol ratio increased to 0.84 g/cm³ at ratios of 1:8 and 1:10. The density value obtained from the esterification of bio-oil meets fuel standards, where the viscosity of diesel ranges from 0.82–0.86 g/cm³ [47, 48]. This decrease in density indicates that esterification produces a lighter, more liquid ester than bio-oil, making it easier to use as a fuel [49, 50].

Diesel viscosity is generally in the range of 1.9–3.8 cSt, while biodiesel viscosity ranges from 3.6–5.0 cSt [51, 52]. Compared with this range, the viscosity values in this study are relatively lower: 0.99 cSt in bio-oil, decreasing to 0.98 cSt after distillation and 0.90 after esterification. This decrease in viscosity is influenced by the increase in the methanol ratio, which indicates the formation of esters with lower viscosity and greater volatility, so that its application suitability requires a fluid with low viscosity, as it is less suitable for use as a single fuel in diesel engines [16, 48]. Therefore, the viscosity of the bio-oil produced has only the potential to be used as an additive or blending component with other fuels to achieve an appropriate viscosity range and ensure lubrication, meeting fuel standards and engine needs. Therefore, optimization and further testing are needed to determine its practical application in fuel formulations.

Among the various physicochemical parameters of fuel, the HHV is a fundamental component that can determine the energy potential of bio-oil. Biofuel resulting from esterification shows an increase, especially at a methanol ratio of 1:10, producing the highest HHV (45.15 MJ/kg), whereas ratios of 1:6 and 1:8 show no significant increase compared to bio-oil (43.55 MJ/kg) or distillate (43.64 MJ/kg). This indicates that distillation-based purification does not significantly increase the HHV. The increase in HHV during esterification results from the conversion of organic acids, such as acetic acid, into esters that are more stable and more efficient in combustion. Research by Aziz et al. [53], on the esterification of bio-oil using sulfuric acid catalysts, reported an HHV of 34.78–41.52 MJ/kg. Meanwhile, research by Khuenkaeo et al. found that esterification of bio-oil with Amberlyst-15 catalyst produced a HHV of 32.2–35.1 MJ/kg [44]. This shows that the HAAZ/MgO catalyst used in the esterification process significantly increases the HHV of the ester. Therefore, the HHV in this study is relatively low compared to conventional liquid fuels such as gasoline (±46 MJ/kg), but still higher than diesel (±43 MJ/kg) and petroleum (±42 MJ/kg) [54], so that the increase in HHV obtained indicates that the processing process can increase the energy content of bio-oil and improve its efficiency as a fuel.

The free organic acid content decreased drastically from 0.019% in bio-oil to 0.01% after the esterification process with methanol (1:6, 1:8 and 1:10). This decrease indicates an increase in stability and resistance to oxidation, the reduction in free organic content indicates that the addition of methanol encourages the conversion of organic acid compounds into esters, thereby contributing to improving the quality of bio-oil and reducing corrosive

properties. Overall, the esterification process with the help of HAAZ/MgO catalysts has a significant effect on the physicochemical properties of bio-oil in increasing the pH value and reducing water content, density, and viscosity, increasing the HHV, so that these changes produce bio-oil with better characteristics to be used as a renewable energy source.

4 Conclusion

The use of a heterogeneous HAAZ/MgO catalyst in the esterification process improves the physicochemical properties of bio-oil and reduces the organic acid content. The combination of HAAZ and MgO provides both Brønsted acid and basic sites, forming a bifunctional catalyst system that facilitates the conversion of organic acids and alcohols into esters. This improvement is reflected in the decrease in acid content and the increase in several physicochemical parameters, including pH and HHV, indicating an enhancement in bio-oil quality. In addition, the heterogeneous HAAZ/MgO catalyst offers advantages, including good stability and ease of separation, which are beneficial for catalyst recovery and reuse. These results demonstrate the potential of the HAAZ/MgO catalyst for upgrading bio-oil derived from biomass waste and contribute to the development of renewable fuel alternatives.

Author Contribution

Conceptualization, methodology, and funding acquisition M.M.; software, validation, and formal analysis A.Z., N.F., and M.J.; investigation, resources, and data curation M.J. and M.M.; writing—original draft preparation, M.M., Y.M., and D.F.; writing—review, editing, and visualization M.M. and Y.M.; project administration D.F. and Y.M. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

References

- [1] Y. Cao and A. Pawlowski, “Biomass as an answer to sustainable energy: Opportunity versus challenge,” *Environ. Prot. Eng.*, vol. 39, no. 1, pp. 5–16, 2013. <https://doi.org/10.5277/EPE130112>
- [2] Y. Milen, M. F. Samawi, E. B. Demmalino, and M. Jahiding, “Calorific value performance of sago dregs biomass products for coal co-firing material,” *J. Ecol. Eng.*, vol. 26, no. 11, pp. 189–203, 2025. <https://doi.org/10.12912/27197050/211942>
- [3] M. Mashuni, L. O. Kadidae, M. Jahiding, F. H. Hamid, W. A. Kadir, and A. M. N. Khaeri, “Chemical characterization and antibacterial activities of bio-oil from durian shell pyrolysis,” *J. Kim. Valensi*, vol. 10, no. 1, pp. 86–96, 2024. <https://doi.org/10.15408/jkv.v10i1.37674>
- [4] N. Sareekam, S. K. Kamarudin, and N. H. Kasmuri, “Optimization of bio oil from palm oil fronds via fast pyrolysis,” *Indian J. Sci. Technol.*, vol. 9, no. 21, pp. 1–13, 2016. <https://doi.org/10.17485/ijst/2016/v9i21/95232>
- [5] Z. E. Zadeh, A. Abdulkhani, O. Aboelazayem, and B. Saha, “Recent insights into lignocellulosic biomass pyrolysis: A critical review on pretreatment, characterization, and products upgrading,” *Processes*, vol. 8, no. 7, p. 799, 2020. <https://doi.org/10.3390/pr8070799>
- [6] A. Bhatnagar, R. Barthen, H. Tolvanen, and J. Konttinen, “Bio-oil stability through stepwise pyrolysis of groundnut shells: Role of chemical composition, alkali and alkaline earth metals, and storage conditions,” *J. Anal. Appl. Pyrolysis*, vol. 157, no. 1, p. 105219, 2021. <https://doi.org/10.1016/j.jaap.2021.105219>
- [7] D. Pfaff, B. M. Jenkins, and S. Q. Turn, “Elemental gas-particle partitioning in fluidized bed combustion and gasification of a biomass fuel,” in *Progress in Thermochemical Biomass Conversion*, 2001. <https://doi.org/10.1002/9780470694954.ch57>
- [8] Y. Yee, S. Thangalazhy-Gopakumar, S. Gan, L. Yee, and H. Kiat, “Esterification and neutralization of bio-oil from palm empty fruit bunch fibre with calcium oxide,” *Bioresour. Technol. Rep.*, vol. 12, no. 6, p. 100560, 2020. <https://doi.org/10.1016/j.biteb.2020.100560>

- [9] L. Zhang, B. Sheng, Z. Xin, Q. Liu, and S. Sun, "Kinetics of transesterification of palm oil and dimethyl carbonate for biodiesel production at the catalysis of heterogeneous base catalyst," *Bioresour. Technol.*, vol. 101, no. 21, pp. 8144–8150, 2010. <https://doi.org/10.1016/j.biortech.2010.05.069>
- [10] M. A. Jamil, "Production and optimization study of biodiesel produced from non-edible seed oil," *Sci. Technol. Energy Transit.*, vol. 79, 2024. <https://doi.org/10.2516/stet/2024036>
- [11] F. Mahmudova, A. Maksumova, N. Matkarimova, S. Tadjieva, and I. Latipova, "Study of the reaction of olefins with acetic acid," *E3S Web of Conferences*, vol. 402, p. 14020, 2023. <https://doi.org/10.1051/e3sconf/202340214020>
- [12] I. Sindhu, R. Tomar, and A. Singh, "Introduction to solid base catalyst," in *Solid Base Catalysts: Synthesis, Characterization, and Applications*. Wiley, 2024. <https://doi.org/10.1002/9783527846719.ch>
- [13] S. Thampy, N. Ashburn, K. Cho, and J. W. P. Hsu, "Earth-abundant transition metal-based mullite-type oxide catalysts for heterogeneous oxidation reactions," *Adv. Energy Sustainability Res.*, vol. 2, no. 6, p. 2000075, 2021. <https://doi.org/10.1002/aesr.202000075>
- [14] E. Bojaddayni, M. E. Küçük, Y. E. Ouardi, I. Jilal, S. E. Barkany, K. Moradi, E. Repo, K. Laatikainen, and A. Ouammou, "A review on synthesis of zeolites from natural clay resources and waste ash: Recent approaches and progress," *Miner. Eng.*, vol. 198, p. 108086, 2023. <https://doi.org/10.1016/j.mineng.2023.108086>
- [15] M. Milina, S. Mitchell, and J. Pérez-Ramírez, "Prospectives for bio-oil upgrading via esterification over zeolite catalysts," *Catal. Today*, vol. 235, pp. 176–183, 2014. <https://doi.org/10.1016/j.cattod.2014.02.047>
- [16] N. Suárez, J. Pérez-Pariente, C. Márquez-Álvarez, and M. Grande, "Preparation of mesoporous beta zeolite by fluoride treatment in liquid phase: Textural, acid, and catalytic properties," *Microporous Mesoporous Mater.*, vol. 284, no. 2, pp. 296–303, 2019. <https://doi.org/10.1016/j.micromeso.2019.04.049>
- [17] M. Jahiding, E. S. Hasan, Y. Milen, Mashuni, and F. S. Purnamasari, "The effect of pyrolysis temperature on bio-oil and bio-char decomposition process of coconut shell for bio-coke hybrid application as alternative energy resources," *AIP Conf. Proc.*, vol. 2973, no. 1, p. 020012, 2024. <https://doi.org/10.1063/5.0184524>
- [18] S. Kadarwati, E. Apriliani, R. N. Annisa, J. Jumaeri, and S. Wahyuni, "Esterification of bio-oil produced from sengon (*Paraserianthes falcataria*) wood using indonesian natural zeolites," *Int. J. Renew. Energy Dev.*, vol. 10, no. 4, pp. 747–754, 2021. <https://doi.org/10.14710/ijred.2021.35970>
- [19] E. N. Ganic and T. G. Hicks, *McGraw-Hill's Engineering Companion*. McGraw-Hill Professional, 2002.
- [20] N. M. Julkapli and S. Bagheri, "Magnesium oxide as a heterogeneous catalyst support," *Rev. Inorg. Chem.*, vol. 36, no. 1, pp. 1–41, 2016. <https://doi.org/10.1515/revic-2015-0010>
- [21] V. Crupi, D. Majolino, F. Longo, P. Migliardo, and V. Venuti, "FTIR/ATR study of water encapsulated in Na-A and Mg-exchanged A-zeolites," *Vib. Spectrosc.*, vol. 42, no. 2, pp. 375–380, 2006. <https://doi.org/10.1016/j.vibspec.2006.05.007>
- [22] R. L. White, "A temperature perturbation infrared spectroscopy comparison of HY and NaY zeolite dehydration/rehydration," *Minerals*, vol. 14, no. 1, p. 104, 2024. <https://doi.org/10.3390/min14010104>
- [23] R. A. Fauzi, D. Sukandar, S. Tursiloadi, A. A. Dwiatmoko, S. Sudiyarmanto, F. Aulia, and N. Rinaldi, "A preliminary study of zeolite-catalyzed esterification of glycerol with fatty acids," *AIP Conf. Proc.*, vol. 2024, no. 1, p. 020045, 2018. <https://doi.org/10.1063/1.5064331>
- [24] P. S. Endang, R. A. Rachman, U. T. I. Martia, A. R. Widyanto, R. M. Iqbal, and N. Widiastuti, "The MnO₂/Zeolite NaY catalyzed oxidation of CO emission in catalytic converter system," *Mater. Sci. Forum*, vol. 964, pp. 199–208, 2019. <https://doi.org/10.4028/www.scientific.net/MSF.964.199>
- [25] J. G. Vitillo, T. Fjermestad, M. D'Amore, M. Milanese, L. Palin, G. Ricchiardi, and S. Bordiga, "On the structure of superbasic (MgO)_n sites solvated in a faujasite zeolite," *Phys. Chem. Chem. Phys.*, vol. 20, no. 27, pp. 18 503–18 514, 2018. <https://doi.org/10.1039/C8CP01788C>
- [26] T. Shan, S. Anto, T. Mathimani, K. Brindhadevi, and A. Pugazhendhi, "Upgrading of bio-oil from thermochemical conversion of various biomass–Mechanism, challenges and opportunities," *Fuel*, vol. 278, p. 119329, 2020. <https://doi.org/10.1016/j.fuel.2020.119329>
- [27] N. M. Marzouk, A. O. A. E. Naga, S. A. Younis, S. A. Shaban, A. M. E. Torgoman, and F. Y. E. Kady, "Process optimization of biodiesel production via esterification of oleic acid using sulfonated hierarchical mesoporous ZSM-5 as an efficient heterogeneous catalyst," *J. Environ. Chem. Eng.*, vol. 9, no. 2, p. 105035, 2021. <https://doi.org/10.1016/j.jece.2021.105035>
- [28] M. Hashemi, K. T. Q. Nguyen, D. J. Robert, G. K. Zhang, T. Hosseinnajad, and D. Marney, "Advanced pyrolysis methods for mixed plastic waste management: Enhanced char and gas yields," *J. Anal. Appl. Pyrolysis*, vol. 193, no. 2, p. 107360, 2026. <https://doi.org/10.1016/j.jaap.2025.107360>
- [29] M. Jahiding, M. Mashuni, F. H. Hamid, W. O. S. Ilmawati, and R. Hamdana, "The production of renewable fuels sago dregs and low-density polyethylene by pyrolysis and its characterization," *Sci. Technol. Indones.*, vol. 9, no. 3, pp. 565–576, 2024. <https://doi.org/10.26554/sti.2024.9.3.565-576>

- [30] B. Malleshham, B. G. Rao, and B. M. Reddy, "Production of biofuel additives by esterification and acetalization of bioglycerol," *C. R. Chim.*, vol. 19, no. 10, pp. 1194–1202, 2016. <https://doi.org/10.1016/j.crci.2015.09.011>
- [31] A. V. Bridgwater, "Review of fast pyrolysis of biomass and product upgrading," *Biomass Bioenergy*, vol. 38, pp. 68–94, 2012. <https://doi.org/10.1016/j.biombioe.2011.01.048>
- [32] R. N. Yanti, Yelmiza, and R. Purnawati, "Karbonisasi limbah biomassa ampas sagu sebagai bahan baku bioenergi," *Wahana For.*, vol. 21, no. 1, pp. 40–46, 2026. <https://doi.org/10.31849/81x7c184>
- [33] J. Keiser, M. P. Brady, J. K. Thomson, R. Connatser, S. Lewis, and D. Leonard, "Bio-oil properties and effects on containment materials," in *Proceedings of the CORROSION 2014*, 2014, pp. 1–12. <https://doi.org/10.5006/C2014-4423>
- [34] X. Guo, S. Wang, Z. Guo, Q. Liu, Z. Luo, and K. Cen, "Pyrolysis characteristics of bio-oil fractions separated by molecular distillation," *Appl. Energy*, vol. 87, no. 9, pp. 2892–2898, 2010. <https://doi.org/10.1016/j.apenergy.2009.10.004>
- [35] M. Kerscher, L. M. Braun, J. H. Jander, M. H. Rausch, T. M. Koller, P. Wasserscheid, and A. Fröba, "Viscosity, interfacial tension, and density of 2-propanol and acetone up to 423 K by surface light scattering and conventional methods," *Int. J. Thermophys.*, vol. 45, no. 8, pp. 1–24, 2024. <https://doi.org/10.1007/s10765-023-03294-z>
- [36] X. Hu, "Stabilization of bio-oil via esterification," in *Chemical Catalysts for Biomass Upgrading*. Wiley, 2019. <https://doi.org/10.1002/9783527814794.ch4>
- [37] S. Liu, Z. Cheng, B. Li, H. Zeng, W. Liang, Y. Luo, Y. Bai, H. Gao, X. Pan, and X. Shu, "Recent advances in biomass-assisted synthesis of hierarchical porous zeolite," *Mater. Today Sustainability*, vol. 27, p. 100917, 2024. <https://doi.org/10.1016/j.mtsust.2024.100917>
- [38] H. Qin, G. Li, H. Yan, X. Feng, Y. Liu, X. Chen, and C. Yang, "Theoretical study on the adsorption and diffusion behavior of methyl oleate catalytic cracking in hierarchical ZSM-5 zeolite," *CIESC J.*, vol. 75, no. 5, pp. 1870–1881, 2024. <https://doi.org/10.11949/0438-1157.20231015>
- [39] R. A. Fauzi, S. Tursiloadi, A. A. Dwiatmoko, F. Aulia, and N. Rinaldi, "Performance of modified natural zeolites by sodium hydroxide treatments in the esterification of glycerol and oleic acid," *J. Kim. Valensi*, vol. 5, no. 11, pp. 236–241, 2019. <https://doi.org/10.15408/jkv.v5i2.9976>
- [40] W. Zhang, H. Wang, Q. Li, Q. Dong, N. Zhao, W. Wei, and Y. Sun, "The mechanism for the synthesis of 1-methoxy-2-propanol from methanol and propylene oxide over magnesium oxide," *Appl. Catal. A*, vol. 294, no. 2, pp. 188–196, 2005. <https://doi.org/10.1016/j.apcata.2005.07.005>
- [41] H. Petitjean, K. Tarasov, F. Delbecq, P. Sautet, J. M. Krafft, P. Bazin, M. C. Paganini, E. Giamello, M. Che, H. Lauron-Pernot, and G. Costentin, "Quantitative investigation of MgO brønsted basicity: DFT, IR, and calorimetry study of methanol adsorption," *J. Phys. Chem. C*, vol. 114, no. 7, pp. 3008–3016, 2010. <https://doi.org/10.1021/jp909354p>
- [42] H. Song, S. Lei, W. Fang, F. Jin, M. Kang, J. Chen, and H. Liu, "Efficient synthesis of tributyl citrate plasticizer via esterification reaction using $\text{SO}_4^{2-}/\text{ZrO}_2\text{--TiO}_2$ as catalysts," *RSC Adv.*, vol. 15, no. 24, pp. 19 339–19 347, 2025. <https://doi.org/10.1039/d5ra02407b>
- [43] H. Ayyub, M. Arslan, M. Jamshaid, A. A. Qureshi, A. Ahmed, H. H. Masjuki, M. A. Kalam, F. B. Ahmad, H. L. Ali, M. U. A. Khan, and et al., "Esterification and transesterification optimization processes of nonedible (castor and neem) oils for the production of biodiesel," *Fuels*, vol. 5, no. 4, pp. 782–802, 2024. <https://doi.org/10.3390/fuels5040043>
- [44] N. Khuenkaeo, P. Prasertpong, and N. Tippayawong, "Upgrading corncob bio-oil via oxidative pretreatment, esterification and azeotropic distillation," *Clean. Eng. Technol.*, vol. 24, no. 10, 2025. <https://doi.org/10.1016/j.clet.2024.100874>
- [45] M. Sahin and M. Ilbas, "Analysis of the effect of H_2O content on combustion behaviours of a biogas fuel," *Int. J. Hydrogen Energy*, vol. 45, no. 5, pp. 3651–3659, 2020. <https://doi.org/10.1016/j.ijhydene.2019.02.042>
- [46] J. Park and J. Oh, "Study on the characteristics of performance, combustion, and emissions for a diesel water emulsion fuel on a combustion visualization engine and a commercial diesel engine," *Fuel*, vol. 311, p. 122520, 2022. <https://doi.org/10.1016/j.fuel.2021.122520>
- [47] M. D. Kass, B. L. Armstrong, B. C. Kaul, R. M. Connatser, S. Lewis, J. R. Keiser, J. Jun, G. Warrington, and D. Sulejmanovic, "Stability, combustion, and compatibility of high-viscosity heavy fuel oil blends with a fast pyrolysis bio-oil," *Energy Fuels*, vol. 34, no. 7, pp. 8403–8413, 2020. <https://doi.org/10.1021/acs.energyfuels.0c00721>
- [48] A. R. K. Gollakota, C. M. Shu, P. K. Sarangi, K. P. Shadangi, S. Rakshit, J. F. Kennedy, V. K. Gupta, and M. Sharma, "Catalytic hydrodeoxygenation of bio-oil and model compounds—Choice of catalysts, and mechanisms, journal = Renew. Sustain. Energy Rev., volume = 187, pages = 113700, year = 2023, doi = 10.1016/j.rser.2023.113700, url = <https://doi.org/10.1016/j.rser.2023.113700>,"

- [49] M. Gulum and A. Bilgin, "An experimental optimization research of methyl and ethyl esters production from safflower oil," *Environ. Clim. Technol.*, vol. 22, pp. 132–148, 2018. <https://doi.org/10.2478/rtuect-2018-0009>
- [50] Y. H. Chan, S. K. Loh, B. L. F. Chin, C. L. Yiin, B. S. How, K. W. Cheah, M. K. Wong, A. C. M. Loy, Y. L. Gwee, S. L. Y. Lo, S. Yusup, and S. Lam, "Fractionation and extraction of bio-oil for production of greener fuel and value-added chemicals: Recent advances and future prospects," *Chem. Eng. J.*, vol. 397, p. 125406, 2020. <https://doi.org/10.1016/j.cej.2020.125406>
- [51] V. S. Shanthini, D. Chitra, and G. Moorthy, "Biodiesel: A comprehensive review of properties, catalyst types, and feedstock sources," *Results Chem.*, vol. 18, p. 102678, 2025. <https://doi.org/10.1016/j.rechem.2025.102678>
- [52] S. Geacai, O. Iulian, and I. Nita, "Measurement, correlation and prediction of biodiesel blends viscosity," *Fuel*, vol. 143, pp. 629–637, 2015. <https://doi.org/10.1016/j.fuel.2014.11.041>
- [53] S. M. A. Aziz, R. Wahi, Z. Ngaini, S. Hamdan, and S. A. Yahaya, "Esterification of microwave pyrolytic oil from palm oil kernel shell," *J. Chem.*, vol. 2017, p. 8359238, 2017. <https://doi.org/10.1155/2017/8359238>
- [54] W. Nor, M. Wan, R. Mamat, H. H. Masjuki, and G. Naja, "Effects of biodiesel from different feedstocks on engine performance and emissions: A review," *Renewable Sustainable Energy Rev.*, vol. 51, pp. 585–602, 2015. <https://doi.org/10.1016/j.rser.2015.06.031>